BayesINE2. Estimación de proporciones. Modelo beta-binomial

Intro a Estadística Bayesiana

DataLab

Abril 2026

ICMAT-CSIC

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Chapter objectives

Objectives

- Introduce main concepts in Bayesian inference through a simple but very useful model: Estimation of proportions through the beta-binomial model
  - Point estimation
  - Interval estimation
  - Hypothesis testing
  - Prediction
  - Decision analysis
- Illustrate strengths and weaknesses
- A lab reinforces concepts and shows robustness of the proposed line of reasoning

Basic setup

Basic setup

Problem faced: Estimating a proportion (no covariates, etc...)

Through a typical example:

Consider fraud detection ML protocols for an administration handling tax exemptions.
We introduce one protocol and wish to assess it, e.g. to compare with another one. Protocol tested in 12 fraudulent applications. Effective in 9 (e.g. no loss induced)

Basic setup

Probabilistic setup

- n trials (identical, independent). Two results: success, failure
- Number X of successes in n trials
- Parameter of interest: θ_1 success probability in a trial
- Distribution number of successes in n trials:
  $X|\theta \sim Bin(n, \theta_1)$
- For the observed $X = 9$, $Pr(X = 9|\theta_1) \propto \theta_1^9(1-\theta_1)^{12-9}$

Basic setup

Classical setup

- Likelihood $l(\theta_1|X=9) \propto \theta_1^9(1-\theta_1)^3, \theta_1 \in [0,1]$
- log-likelihood $h(\theta) \propto 9 \times \theta_1 + 12 \times (1-\theta_1), \theta_1 \in [0,1]$
- MLE. Maximum likelihood estimator
  - Maximize log-likelihood
  - In this example, $\hat{\theta}_1 = 9/12$

Basic setup

Bayesian setup. Another source of uncertainty (or info)

- Parameter viewed as an uncertain quantity
- To fix ideas suppose $p(\theta_1) = 1, \theta_1 \in [0, 1]$
- Any intuition of what we are modeling?

Bayesian setup. We need to aggregate both sources of information: Bayes' formula. From the prior to the posterior

- $p(\theta_1|X = 9) \propto p(\theta_1) \times p(X = 9|\theta_1) \propto \theta_1^9(1 - \theta_1)^3, \theta_1 \in [0, 1]$

The posterior summarizes all information available about θ_1. In this case, a $Beta(10, 4)$ distribution

Basic setup

Bayesian setup. Sequential update: Today's posterior serves as prior for tomorrow.

E.g. if in next 5 applications, 3 successes

$p(\theta_1|X=3) \propto [\theta_1^9(1-\theta_1)^3] \times [\theta_1^3(1-\theta_1)^2], \theta_1 = [\theta_1^{12}(1-\theta_1)^5]$

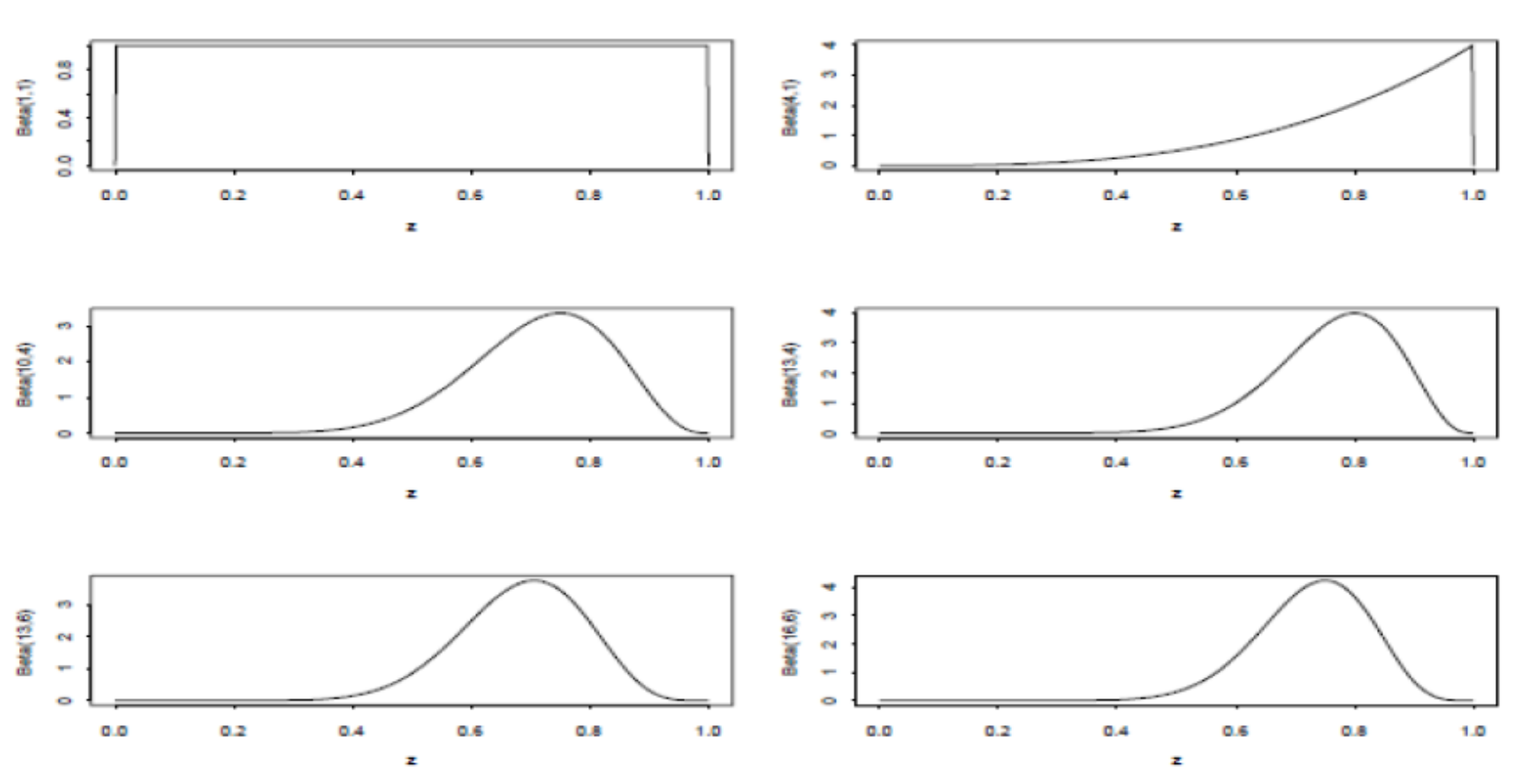
corresponding to a $Beta(13,6)$

The posterior clearly depends on the prior. Suppose that a priori, the success probability is around 80% and bigger values are deemed more likely. The learning process goes through

$$Beta(4,1) \rightarrow Beta(13,4) \rightarrow Beta(16,6$$

Basic setup

Both cases side by side:



Convergence of learning, Consensus, Asymptotic behaviour

Basic uses. Inference and prediction

Basic uses

Posterior summarises all info available to perform

- Point estimation
- Interval estimation
- Hypothesis testing
- Prediction
- Decision analysis

Focus on the case Beta (10,4). 'Analytical' results.

Final recap for general procedures.

Use simulation for all computations also in Lab 2!!!

Basic uses. Point estimation

Summarise the posterior in a single θ_1 value

E.g. the posterior mean $\frac{10}{10+4} = 0.72$

Other possibilities: Posterior median, posterior mode (MAP),...

Choice based on minimising expected loss

Basic uses. Interval estimation

Summarise the posterior in an interval with a desired probability level (of containing θ_1)

E.g Symmetric interval of probability 0.8
$[qbeta(0.1, 10, 4), qbeta[0.9, 10, 4)] = [0.55, 0.86]$

Other possibilities: Higher posterior density (HPD) Interval,...

Choice based on minimising expected loss...

Compare with confidence intervals...

Basic uses. Hypothesis testing

Which among several hypothesis supported by the evidence?

- E.g. Is the protocol effective? (aka is $\theta_1 \geq 0.5$)
- E.g Select the hypothesis with highest posterior probability $1 - pbeta(0.5, 10, 4) = 0.953$.

Many other criteria. Choice based on minimising expected loss...

Basic uses. Forecasting

Forecasting results in future observations.

Probability of more than 4 successes in next 7 trials?

The predictive distribution.

$$
\begin{aligned}
Pr(X = k|x = 9) &= \int Pr(X = k|\theta_1)p(\theta_1|x = 9)d\theta_1 = \\
&= \int \binom{7}{k}\theta_1^k(1-\theta_1)^{7-k}\binom{13}{3}\theta_1^9(1-\theta_1)^3 d\theta_1 = \\
&= \frac{\binom{7}{k}\binom{13}{3}}{\binom{20}{9+k}}.
\end{aligned}
$$

$$
Pr(X \geq 5|x = 9) = \sum_{k=5}^{7} Pr(X = k|x = 9) = 0.6641.
$$

Bayesian Decision Analysis

Bayesian decision analysis

Support (even automate) decision making

Choose among several decisions given evidence available....

And preference over consequences

Through maximum expected utility (or minimum expected loss)

Posterior expected utility

Predictive expected utility

Bayesian decision analysis

Consider a second protocol. 10 opportunities, successful in 6.
Want $r = Pr(\theta_1 \geq \theta_2 | X = 9, Y = 6)$

Model is

$$X|\theta_1 \sim Bin(12, \theta_1)$$

$$Y|\theta_2 \sim Bin(10, \theta_2)$$

$$\theta_1, \theta_2 \sim Unif[0, 1]$$

independent

Bayesian decision analysis

$$\theta_1 \sim Beta(10, 4), \theta_2 \sim Beta(7, 5)$$

- Distribution of $\theta_1 - \theta_2$????
- Through simulation. E.g 1000 observations, compute differences, count those bigger than 0, divide by 1000.
- Which protocol is better?

$$r \approx 0.772.$$

Bayesian decision analysis

- Utility structure

	succeeds	**does not succeed**
Plan A	0.8	0
Plan B	1	0.2

- Expected utilities given probabilities

$$0.8\theta_1 + 0(1-\theta_1) = 0.8\theta_1 \qquad \theta_2 + 0.2(1-\theta_2) = 0.2 + 0.8\theta_2$$

- Expected utilities

$$0.8E(\theta_1|x=9) = 0.8 \times \frac{10}{14} = \frac{4}{7} \quad 0.2 + 0.8E(\theta_2|y=6) = 0.2 + 0.8 \times \frac{7}{12} = \frac{2}{3}$$

Recap

Recap. Proportion estimation

Classical setup

- Parameter of interest: θ success probability in a trial
- Model Number X of successes in n trials $X|\theta \sim Bin(n, \theta)$
- $Pr(X = x|\theta_1) = \binom{n}{x}\theta^x(1-\theta)^{n-x}$
- Data x
- Likelihood $l(\theta|x) \propto \theta^x(1-\theta)^{n-x}$
- MLE
  - $h(\theta) \propto \theta^x(1-\theta_1)^{n-x}, \theta_1 \in [0,1]$
  - $h'(\theta) = 0 \implies \hat{\theta} = \frac{x}{n}$

Recap. Proportion estimation

Bayesian setup

- Parameter of interest: θ, Binomial model
- Data x, Likelihood $l(\theta|x) \propto \theta^x(1-\theta)^{n-x}$
- Prior Beta (a,b)
  - $p(\theta) \propto \theta^{a-1}(1-\theta_1)^{\beta-1}, \theta \in [0,1]$
- Posterior Beta (a+x,b+n-x)
  - $p(\theta) \propto \theta^{a-1}(1-\theta_1)^{\beta-1} \times \theta^x(1-\theta_1)^{n-x}\theta \in [0,1]$
  - Sequential update ...

Recap. Proportion estimation

Bayesian point estimation

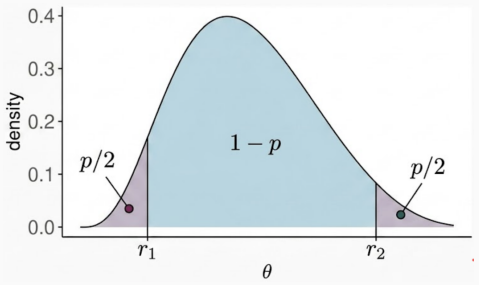
Posterior Beta (a+x,b+n-x)

- Posterior mean $E(\theta|x) = \frac{a+x}{a+b+n}$
  - $\frac{a+x}{a+b+n} = \frac{n}{a+b+n}\frac{x}{n} + \frac{a+b}{a+b+n}\frac{a}{a+b}$
  - As $n \to \infty$, $E(\theta|x) \approx \frac{x}{n}$, $Var(\theta|x) \to 0$.
- Posterior median $q(0.5, a+x, b+n-x)$
- Posterior mode $\frac{a+x-1}{a+b+n-2}$
- ...
- Laplace smoothing

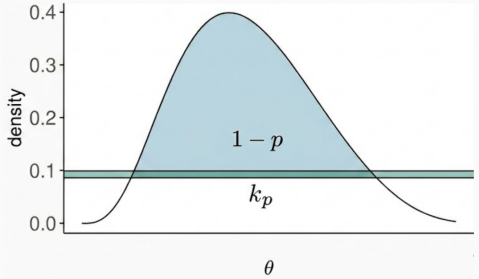
Recap. Proportion estimation

Bayesian interval estimation Posterior Beta (a+x,b+n-x)

- Symmetric Interval
  $[qbeta(p/2, a+x, b+n-x), qbeta(1-p/2, a+x, b+n-x)]$



- Highest Posterior Density interval
  $Pr\{\theta : p(\theta|x) \geq k_p\} \geq (1-p)$



- ....

Recap. Proportion estimation

Hypothesis testing Posterior Beta (a+x,b+n-x)

- Testing three hypothesis $\Theta_1, \Theta_2, \Theta_3$
  - Compute $Pr(\Theta_i|x)$
  - Choose hypothesis with biggest posterior probability
  - 0-1 loss or 1-0 utility
- Testing a point null. $\theta = \theta_0$ vs $\theta \neq \theta_0$
  - Sensible problem?
  - Practical solution
    - Build credible interval R for θ
    - Evidence supports null if $\theta_0 \in R$
- ....

Recap. Proportion estimation

Forecasting successes in m future trials

Posterior Beta (a+x,b+n-x)

Predictive distribution

$Pr(Y = k|X = x) = \int Pr(Y = k|\theta)p(\theta|x)d\theta$

$= \frac{\binom{m}{k}\beta(a+x)\beta(b+(n-x))}{\beta(k+a-x)\beta((m-k)+b+(n-x))}$

Summarising the predictive distribution E.g.

$E(Y|x) = \sum kPr(Y = k|x) = m\frac{a+x}{b+(n-x)}$

...

Recap

The reasoning is general

- Simpler interpretations
- More natural concepts
- Probability distributions for observations and parameters
- Integration of expert opinion
- Apportioning of all sources of uncertaimty
- Integration with decision making

Final comments

Final comments

- We have used conjugate priors. The beta is a conjugate prior of the binomial (the posterior is also beta). But this is available only in simpler models.
- Assessing the prior
  - Noninformative prior. Uniform (Beta (1,1)), Jeffrey's (Beta (1/2, 1/2)),...
  - Assess a couple of quantiles and deduce a and b
  - Observables
- Consistency
- Computations

References

References

Agresti, Kateri, Grove, Mira (2026) Bayesian Statistics: A primer for data scientists with Python and R.

Gelman, Carlin, et al (2014) Bayesian Data Analysis.

French, Ríos Insua (2000) Statistical Decision Theory.

Lab2

Lab2

The beta-binomial model in R

The gamma-exponential model in R